1) Given:

Process	AT	BT	Priority
P1	0	12	2
P2	1	5	6
P3	10	10	1
P4	5	18	4
P5	9	3	3
P6	3	1	5

(i) FCFS - Without Arrival Time:

P1	P2	P3	P4	P5	P6

0 12 17 27 45 48 49

P_id	AT	BT	FT	WT	TAT
P1	0	12	12	0	12
P2	1	5	17	11	16
P3	10	10	27	17	17
P4	5	18	45	22	40
P5	9	3	48	36	39
P6	3	1	49	45	46

Avg WT = 24.83 (or) 24.83

Avg TAT = 28.33 (or) 33

(ii) FCFS - With AT

P1	P2	P6	P4	P5	P3

0 12 17 18 36 39 49

P_id	AT	BT	FT	WT	TAT
P1	0	12	12	0	12
P2	1	5	17	11	16
P3	10	10	49	29	39
P4	5	18	36	13	33
P5	9	3	39	27	30
P6	3	1	18	14	15

Avg WT = 15.67

Avg TAT = 23.83

(iii) SJF - Without AT

P_6	P_5	P_2	P_3	P_1	P_4

0 1 4 9 19 31 49

P_id	AT	BT	FT	WT	TAT
P1	0	12	31	19	31
P2	1	5	9	4	9
P3	10	10	19	19	19
P4	5	18	49	31	49
P5	9	3	4	1	4
P6	3	1	1	0	1

Avg WT = 15.67

Avg TAT = 23.83

(iv) SJF- preemptive

P1	P2	P6	P2	P1	P5	P1	P3	P4

0 1 3 4 7 9 12 21 31 49

P_id	AT	BT	FT	WT	TAT
P1	0	12	21	9	21
P2	1	5	7	1	6
P3	10	10	31	11	1
P4	5	18	49	26	3
P5	9	3	12	0	21
P6	3	1	4	0	44

Avg WT = 7.83

Avg TAT = 16

(v) SJF - Non-preemptive

P1	P2	P6	P5	P3	P4

0 12 17 18 21 31 49

P_id	AT	BT	FT	WT	TAT
P1	0	12	12	0	12
P2	1	5	17	11	16
P3	10	10	31	21	21
P4	5	18	49	26	44
P5	9	3	21	9	12
P6	3	1	18	14	15

Avg WT = 10.83

Avg TAT = 19

(vi) Priority - without AT

P3	P1	P5	P4	P6	P2

0 10 22 25 43 44 49

P_id	AT	BT	FT	WT	TAT
P1	0	12	22	10	22
P2	1	5	49	44	49
P3	10	10	10	0	10
P4	5	18	43	25	43
P5	9	3	25	22	25
P6	3	1	44	43	44

Avg WT = 24

Avg TAT = 32.17

(vii) Priority preemptive

P1	P5	P3	P1	P5	P4	P6	P2

0 9 10 20 28 30 48 49 54

P_id	AT	BT	FT	WT	TAT
P1	0	12	28	16	28
P2	1	5	54	48	53
P3	10	10	20	0	10
P4	5	18	48	25	43
P5	9	3	30	18	21
P6	3	1	49	45	46

Avg WT = 25.33

Avg TAT = 33.5

(viii) Priority - Non preemptive :

P1	P3	P5	P4	P6	P2

0 12 22 25 43 44 49

P-id	AT	BT	FT	TAT	WT
P1	0	12	12	12	0
P2	1	5	49	48	43
P3	10	10	22	12	2
P4	5	18	43	38	20
P5	9	3	25	16	13
P6	3	1	44	41	40

Avg WT = 19.67

Avg TAT = 27.83

(ix) Round Robin (Time quantum = 3)

~~P1 P2 P6 P4 P5 P3 P1 P4 P3 P4~~

~~0 3 6 7 10 13 16 19 22 25 49~~

P1	P2	P6	P4	P5	P1	P2	P3	P4	P1	P3	P4	P1	P3	P4	P4

0 3 6 7 10 13 16 18 21 24 27 30 33 36 39 42 43

P4	P4

43 46 49

P	AT	BT	FT	TAT	WT
P1	0	12	36	36	24
P2	1	5	18	17	12
P6	3	1	7	4	3
P5	9	3	13	4	1
P3	10	10	43	33	23
P4	5	18	49	44	26

Avg WT = 14.83

Avg TAT = 23

2) Given :

System	AT	BT	priority
A	0	5	1
B	1	2	2
C	2	10	1
D	3	3	3

* FCFS

A	B	C	D

0 5 7 17 20

P_id	AT	BT	FT	WT	TAT
A	0	5	5	0	5
B	1	2	7	4	6
C	2	10	17	5	15
D	3	3	20	14	17

Avg WT = 5.75

Avg TAT = 10.75

* SJF - Non-preemptive

A	B	D	C

0 5 7 10 20

P_id	AT	BT	FT	WT	TAT
A	0	5	5	0	5
B	1	2	7	4	6
D	3	3	10	4	7
C	2	10	20	8	18

Avg WT = 4

Avg TAT = 9

* SJF - preemptive

A	B	D	C

0 5 7 10 20

Avg WT = 4

Avg TAT = 9

* Priority - Non-preemptive

A	C	B	D

0 5 15 17 20

P_id	AT	BT	FT	WT	TAT
A	0	5	5	0	5
C	2	10	15	3	15
B	1	2	17	14	17
D	3	3	20	14	20

Avg WT = 7.75

Avg TAT = 12.75

* Priority - Preemptive

A	C	B	D

0 5 15 17 20

P	AT	BT	FT	WT	TAT
A	0	5	5	0	5
B	1	2	17	14	16
C	2	10	15	3	13
D	3	3	20	14	17

Avg WT = 7.75

Avg TAT = 12.75

* Round Robin (Time quantum = 1)

A	B	C	D	A	B	C	D	A	C	D	A	C	A	C	C	C	C	C	C

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Process	AT	BT	FT	WT	TAT
A	0	5	14	9	14
B	1	2	6	3	5
C	2	10	20	8	18
D	3	3	8	2	5

Avg WT = 5.5

Avg TAT = 10.5

∴ SJF (Non-preemptive) is Best overall.